

Python Collections Interview Questions with Detailed Answers

1. What is a list in Python?

A list is an ordered and mutable collection that allows duplicate values. Lists are created using square brackets []. Example: numbers = [1, 2, 3].

2. How is a list different from a tuple?

Lists are mutable and use []. Tuples are immutable and use (). Lists are slower compared to tuples.

3. Are lists mutable or immutable?

Lists are mutable, meaning their elements can be changed after creation.

4. How do you remove duplicates from a list?

You can use set() to remove duplicates: list(set(a)). To preserve order: list(dict.fromkeys(a)).

5. What is list comprehension?

List comprehension is a concise way to create lists in a single line. Example: squares = [x*x for x in range(5)].

6. Difference between append() and extend()?

append() adds a single element to the end of the list. extend() adds multiple elements individually.

7. What is the difference between insert() and append()?

append() adds element at the end. insert(index, value) adds element at a specific position.

8. What is a tuple?

A tuple is an ordered and immutable collection that allows duplicate values.

9. Why are tuples immutable?

Tuples are immutable to ensure data safety and to allow them to be used as dictionary keys.

10. Why are tuples faster than lists?

Tuples are faster because they are immutable and require less memory overhead.

11. What is tuple unpacking?

Tuple unpacking assigns tuple values to multiple variables at once. Example: `a, b = (1, 2)`.

12. What is packing and unpacking in tuples?

Packing: `t = 1,2,3`. Unpacking: `a,b,c = t`.

13. How do you convert a tuple to a list?

Use `list(tuple_name)`. Example: `l = list(t)`.

14. What is a set in Python?

A set is an unordered collection that does not allow duplicate values.

15. Why does a set not allow duplicate values?

Sets use hashing internally, so each element must be unique.

16. Are sets ordered or unordered?

Sets are unordered collections.

17. Can we access elements using index in a set?

No, sets do not support indexing.

18. What is a dictionary in Python?

A dictionary stores data as key-value pairs. Example: `d = {'name': 'John', 'age': 25}`.

19. What types can be used as dictionary keys?

Dictionary keys must be immutable types such as int, float, string, or tuple.

20. Why must dictionary keys be immutable?

Because dictionaries use hashing for fast lookups, and hash values must not change.

21. Difference between `get()` and `[]` in dictionaries?

Using `[]` raises `KeyError` if key is missing. `get()` returns `None` or a default value instead of raising error.

22. What is the difference between `items()`, `keys()`, and `values()`?

`keys()` returns all keys, `values()` returns all values, `items()` returns key-value pairs.

23. How do you remove a key from a dictionary?

Use `del d[key]` or `d.pop(key)`.

24. What is the difference between `pop()` and `del`?

`pop()` removes and returns the value. `del` removes the key but does not return the value.

25. What are set operations?

Union ($A \cup B$) returns all elements. Intersection ($A \cap B$) returns common elements. Difference ($A - B$) returns elements in A not in B. Symmetric Difference ($A \oplus B$) returns elements in either A or B but not both.